

Sentinel-1 GRD
IW, 10 m, VV+VH (2014–2021)

Preprocessing
incidence-angle orbit selection ($\geq 90\%$ coverage)
focal-mean speckle filter

VV Otsu
per-scene threshold
single-pol

Dual-pol SVM
per-scene retrain
VV+VH

Automated training
JRC water occ. $\geq 95\%$
WorldCover land

Identical post-processing
gap-fill $\rightarrow$ vectorise $\rightarrow$ keep in-reservoir polygon

Surface-area series + morphometry
dynamic A/P per scene ·
LOWESS temporal smoothing

Validation & cost
JRC (62) · PlanetScope 3 m (4) · EECU audit